MIL-P-47067(MI)
10 May 1974
SUPERSEDING
MIS 16415A
17 June 1969

MILITARY SPECIFICATION

2, 4-PENTANEDIONE

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of 2, 4-pentanedione.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-414	Sampling Procedures and Tables for Inspection By Variables for Percent Defective

FSC 6810

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on the date of invitation for bids or requests for proposal shall apply.

Uniform Classification Committee

Uniform Freight Classification Rules

(Application for copies should be addressed to Uniform Classification Committee, 202 Chicago Union Station, Chicago, Illinois 60606.)

American Trucking Association Publication

National Motor Freight Classification Rules and Container Regulations

(Application for copies should be addressed to the National Classification Board, 1424 Sixteenth Street, N.W., Washington, D.C. 20013.)

3. REQUIREMENTS

3.1 Properties. The material shall conform to the following properties:

		Min	Max
a.	Assay, 2, 4-pentanedione, weight percent	95.0	----
b.	Specific gravity, 20/20 degrees Celsius (C)	0.972	0.978

3.2 Workmanship. The material shall be free from all impurities and other defects that could adversely affect its performance. The material shall be uniform in quality and shall be manufactured in accordance with standard manufacturing procedures of the industry.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot size. A lot shall be as defined in MIL-STD-414 with the following additional requirements: if manufacture is by batch process, each batch, blend of batches, or fraction thereof, shall constitute a lot.

4.3 Sampling. Sampling shall be in accordance with MIL-STD-414, Inspection Level III, Sections A and C, at the acceptable quality levels shown in Table I for the lower specification limits (LSL) and upper specification limits (USL).

TABLE I

Quality Conformance Test Schedule			
Characteristic	Acceptable Quality Levels (%)		Test Method
	(LSL)	(USL)	
a. Assay	1.0	----	4.4.1
b. Specific gravity	1.5	1.5	4.4.2

4.4 Quality conformance tests and methods. Quality conformance tests and test methods shall be as specified in Table I.

4.4.1 Assay. The assay of the material shall be determined as follows:

a. Dissolve 25 grams (g) of cupric acetate mono-hydrate and 100 g of reagent-grade sodium acetate trihydrate in sufficient water to make one liter of solution. Filter before using.

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b. Pipet 100 milliliters (ml) of the acetate solution into a 500-ml iodine flask containing 75 ml distilled water. Reserve this solution as a blank for the procedure described in (t).

c. Pipet 100 ml of the acetate solution into a 250 ml glass-stoppered flask.

d. Add 1.2000 to 1.5000 g of the sample by means of a weighing pipet.

e. Allow the flask to stand for five minutes with occasional shaking.

f. Fit a 50 ml sintered-glass filter of medium porosity into the top of a 500 ml pear-shaped separatory funnel having a suitable stopcock sealed to the shoulder of the funnel. Apply vacuum to the funnel through this stopcock. Close the stopcock at the bottom of the funnel.

g. Carefully add the contents of the flask to the filter and allow the filtrate to drain into the funnel.

h. Rinse the inside walls of the flask with four 5-ml portions of distilled water, using the rinsings to wash the precipitate deposited on the filter. Allow each washing to pass completely through the filter before adding the next washing.

i. Close the side-arm stopcock, disconnect the vacuum line, and remove the filter.

j. Add 20 ml of reagent-grade chloroform to the separatory funnel, replace the stopper, invert, and shake vigorously with frequent venting for one minute.

k. Allow the layers to separate and draw off the chloroform layer into a 250-ml separatory funnel containing 40 ml of distilled water.

l. Stopper the funnel, invert, and shake vigorously with frequent venting for 30 seconds.

m. Allow the layers to separate and discard the lower layer.

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n. Add 20 ml of the chloroform to the 500-ml funnel, replace the stopper, invert, and shake vigorously for 30 seconds.

o. Allow the layers to separate and draw off the lower layer into the 250-ml funnel containing the upper layer from (m).

p. Stopper this funnel, invert, and shake vigorously for 30 seconds.

q. Allow the layers to separate and discard the lower layer.

r. Repeat the extractions with additional 20-ml portions of the chloroform, using the procedure described in (a) to (q) until the chloroform from the last extraction is colorless.

s. Transfer the upper layer from both separatory funnels to a 500-ml iodine flask. Rinse each funnel with three 5-ml portions of distilled water and add the rinsings to the flask.

t. Add 15 ml of six normal (N) hydrochloric acid and 8 g of chemically-pure potassium iodine crystals to both this flask and the blank prepared in (b).

u. Replace the stoppers, seal with water, and allow the flasks and contents to stand for 10 minutes with occasional swirling.

v. Titrate the contents of each flask with standard 0.1N sodium thiosulfate to a light brown color. Add 10 ml of a freshly prepared 1.0 percent starch solution and continue the titration until the blue color just disappears.

w. Calculation:

Weight percent 2, 4-pentanedione =

$$\frac{(B-A)\ (N)\ (20.02)}{\text{sample weight}}$$

Where: A = Sodium thiosulfate required for sample, ml

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B = Sodium thiosulfate required for the blank, ml

N = Normality of sidium thiosulfate solution

4.4.2 Specific gravity. The specific gravity of the material shall be determined with a Westphal specific gravity balance in accordance with the following procedure.

a. Calbirate the balance at the test temperature by the following procedure. Fill the sample graduate with freshly boiled, cool, distilled water. Clamp the graduate in place in a constant temperature bath set at 20 plus or minus 10 degrees C. Allow 15 minutes conditioning time to achieve temperature equilibrium. Remove the sample graduate from the bath and place it on the balance. Insert the plumb bob in the distilled water. Adjust the rider and vernier scale on the balance to give a reading of 1.000. This is the specific gravity of distilled water at the test temperature.

b. Fill a sample graduate with the liquid to be tested. Clamp the graduate in place in the constant temperature bath and condition the sample for 15 minutes.

c. Remove the graduate from the bath and place it on the balance and read the balance when it reaches equilibrium. Perform this step as rapidly as possible to minimize temperature changes. For accurate work, the plumb bob should be immersed to the same depth for each determination.

d. Record the balance reading as the specific gravity of test sample at the conditioning temperature. Duplicate results must check within plus or minus 0.002 unit or additional samples must be run.

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4.5 Examination. Packaging and marking of the container shall be inspected for conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Material shall be preserved and packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from supply source to first receiving activity (see 6.2).

5.2 Packing. Packaged material shall be secured and packed in shipping containers in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate. Containers shall comply with Uniform Freight Classification Rules or National Motor Freight Rules as applicable.

5.3 Marking. Containers shall be marked in accordance with MIL-STD-129. Marking shall include, but not be limited to, the following information:

a. Manufacturer's name.

b. Material trade name and chemical name.

c. Lot number.

d. Date of manufacture.

e. Number and revision letter of this specification.

f. Storage requirements.

g. Toxicity precautions, if required.

h. Net weight.

i. Fire prevention precautions.

6. NOTES

6.1 Intended use. The material is intended for use in solid propellants for rocket motors.

6.2 Ordering data. Procurement documents should specify, but not be limited to the following information:

a. Number and revision letter of this specification.

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b. Requirements for preservation, packaging, packing, and marking, if different (see 5.1).

c. Place of inspection.

d. Minimum lot size, if applicable.

e. Place of delivery.

f. Request for three copies of laboratory analysis, if inspected by supply source.

g. Request for unusual storage requirements and toxicity.

6.3 Supersession data. This specification includes the requirements of MIS-16415A, dated 17 June 1969.

Custodian:
ARMY-MI

Preparing Activity:
ARMY - MI
Project No. 6810-AA14

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-47067	2, 4 - Pentanedione

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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